Math 152 Problem Set 3

Due 10/21/2026, at 10:00 PM

Problem 1. Evaluate the indefinite integral

$$\int (1+x)^2 \mathrm{d}x$$

in two ways. First, expand $(1+x)^2$ as a polynomial, and integrate it term by term. Then, use the substitution $u = 1 + x$. Explain why the answers you get agree.

Problem 2. Use substitution to evaluate the following definite integral.[1]

$$\int_0^{\pi/2} \frac{\sin(2x)}{(4+\sqrt{1-\cos(2x)})^3} \mathrm{d}x$$

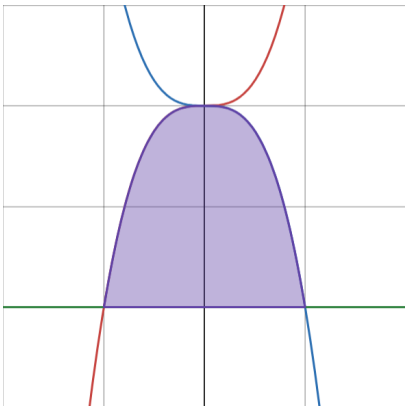
Problem 3. Compute the area of the pictured region enclosed by the curves

$$y = 2x^3 + 3$$

$$y = -2x^3 + 3$$

$$y = 1$$

using an integral in y.



Problem 4. [2] Use the change of variable formula to show that for any $a, b > 0$, we have

$$\int_1^a x^{-1} \mathrm{d}x = \int_b^{ab} x^{-1} \mathrm{d}x.$$

Use this to argue that for $a, b > 0$

$$\int_1^{ab} x^{-1} \mathrm{d}x = \int_1^a x^{-1} \mathrm{d}x + \int_1^b x^{-1} \mathrm{d}x.$$

[1] It is recommended to use multiple substitutions.

[2] In this problem, you should not appeal to any properties of the logarithm function. In fact, you should not use an antiderivative of x^{-1} at all.

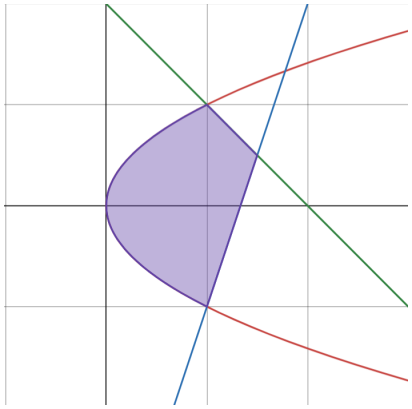
Problem 5. Compute the area of the pictured region enclosed by the curves

$$x = y^2$$

$$y = 2x - 3$$

$$x + y = 2$$

by any method. You do not need to fully simplify the answer.



Problem 6 (Optional)**.** Let f be a continuous function. Argue that for any numbers a, b, c the function

$$G(x) = \int_a^x f(\sin(bx + c)) \cos(bx + c) \mathrm{d}x$$

is periodic. What is its period?

Review Problems

The first midterm will cover material up to Wednesday 10/14, including the computation of areas between curves by integration in x. The following problems cover this material.

Problem. Consider the limit $\lim_{x \to 2}(-3x + 2)$.

(a) What is the value of this limit?

(b) Write down the ε, δ definition in this case.

(c) Suppose you are given $\varepsilon = 1/2$. What is a choice of δ that works?

Problem. Evaluate the following limit.

$$\lim_{t \to 1} \frac{t^3 + 2t^2 - 3t}{t - 1}$$

Problem. Differentiate the following function.

$$f(x) = \frac{1}{1 + \sqrt{2x^2 + 1}}$$

Problem. Consider the definite integral below.

$$\int_1^4 x^2 \mathrm{d}x$$

Use the partition $\mathcal{P} = \{1 < 2 < 3 < 4\}$, which splits the interval $[1, 4]$ into sub intervals $[1, 2]$, $[2, 3]$, and $[3, 4]$, to find an upper bound on the integral.

Problem. Find an antiderivative F of $f(x) = 2\cos(x)$ which satisfies $F(\pi) = -1$.

Problem. Evaluate the following definite integral:

$$\int_{-1}^1 [x^3 - 2x^2 + 4] \mathrm{d}x$$

Problem. Evaluate the following indefinite integral:

$$\int [3x^2 - \cos(x)] \mathrm{d}x$$

Problem. Evaluate the following indefinite integral:

$$\int (x+1)\sin(3(x+1)^2) \mathrm{d}x$$

Problem. Evaluate the following definite integral:

$$\int_{-\pi/2}^{\pi/2} \frac{\cos(x)}{\sqrt{2 + 2\sin(x)}} \mathrm{d}x$$

Problem. Find the area above the curve $y = x^2 - 3$ and below the line $-x - 1$.